

Modularity Theorem for Elliptic Curves

MA4092 Mathematics Seminar II

Fadhlannafis Khawarizmi Kesumajana (10122040)

First Advisor: Rizal Afgani, Ph.D.

Second Advisor: Oki Neswan, M.Sc., Ph.D.

First Examiner: Dr. Math. Fajar Yuliawan, S.T., M.Si.

Second Examiner: Prof. Dr. Muchtadi Intan Detiena, S.Si., M.Si

Undergraduate Program in Mathematics
Faculty of Mathematics and Natural Sciences
Institut Teknologi Bandung

7 July 2026

Contents

- ① Review & Motivation
- ② Schemes, Continued
- ③ Algebro-Geometric Toolbox
- ④ Elliptic Curves
- ⑤ Moduli of Elliptic Curves
- ⑥ Ending

Review & Motivation

Motivation: Modularity Theorem

- We shall do things a bit differently from now on: we start with the main result of this seminar, that we will not prove (due to the proof being long and requiring objects that are beyond the scope of this seminar).

Theorem (Modularity)

Every elliptic curve E over $\mathbb{Q}$ is modular.

- This theorem has the following famous corollary.

Corollary (Fermat's Last Theorem)

Let $n \geq 3$ be an integer. The only solution to the following Diophantine equation

$$a^n + b^n = c^n$$

satisfy $abc = 0$.

Motivation: Modularity Theorem

- To prove Fermat's Last Theorem, we need the following theorem.

Theorem (Ribet's Theorem)

If there exists integers a, b, c such that $a^n + b^n = c^n$ for $n \geq 3$, then the Frey curve $y^2 = x(x - a^n)(x + b^n)$ cannot be modular.

Proof Sketch.

For the sake of contradiction assume that there exist integers a, b, c such that $a^n + b^n = c^n$ for $n \geq 3$. Without loss of generality we can assume that n is prime. Consider the Frey curve $y^2 = x(x - a^n)(x + b^n)$, which is an elliptic curve. By the Modularity Theorem, this curve is modular. However, by Ribet's Theorem, Frey curves cannot be modular, a contradiction. □

Motivation: Modularity Theorem

- In 1995, Andrew Wiles and Richard Taylor proved Fermat's Last Theorem by first proving some subcase of the Modularity Theorem.
- Specifically, they proved a subcase of the following version:

Theorem (Modularity, Galois Representation Version)

Let E be an elliptic curve over $\mathbb{Q}$. Then the Galois representation of E , $\rho_{E,\ell}$, is modular for some prime ℓ (Diamond and Shurman, 2006).

- The proof relies on **Galois deformations**, i.e. deformations of Galois representations.
- More specifically, the strategy relies on proving that the universal deformation ring of Galois representations, R , is isomorphic to a Hecke algebra, $\mathbb{T}$.

Motivation: Modularity Theorem

- Galois representations of elliptic curves are constructed using Tate modules.
- More specifically, the Galois representation of an elliptic curve E is defined as a map $\rho_{E,p} : \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \text{GL}_2(\mathbb{Q}_p)$, since the automorphism group of the Tate module is isomorphic to $\text{GL}_2(\mathbb{Q}_p)$.

Motivation: Modularity Theorem

- Of course, the preceding discussion leaves us with a few questions.
- Namely, what are elliptic curves? What is a "Galois representation"? What is a Hecke algebra? What does "modular" mean? What are modular forms? What are "deformations"?
- Some of these questions, and some prerequisites, shall be answered shortly (hopefully).

Review: Problem Statement and Goals

- Problem Statement: What are the tools needed to prove the Modularity Theorem?
- Goals:
 - ① Understand the basic objects of algebraic geometry.
 - ② Understand a more general version of the definition of elliptic curves.
 - ③ Compare the properties of classical elliptic curves as varieties with its more general version.
 - ④ Identify the role of scheme theory in the proof of modularity theorem.

Review: Basic Algebraic Geometry

- To construct schemes, we need a generalization of the notion "function space".

Definition

A **presheaf** $\mathcal{F}$ over X is for every open set $U \subseteq X$, a set/ring/group/etc $\mathcal{F}(U)$, and for every $U \subseteq V$ open, a mapping $\text{res}_U^V : \mathcal{F}(V) \rightarrow \mathcal{F}(U)$ called the **restriction map** such that:

- ① $\text{res}_U^U = \text{id}_{\mathcal{F}(U)}$ for every $U \subseteq X$ open,
- ② $\text{res}_U^W = \text{res}_U^V \circ \text{res}_V^W$ for $U \subseteq V \subseteq W$ open.

Let X be a topological space and $\mathcal{F}$ be a presheaf on X . We call $\mathcal{F}$ a **sheaf** over X if for $S \subseteq X$ open, $(S_i)_i$ an open cover of S ,

- ① If $s, s' \in \mathcal{F}(S)$ such that $s|_{S_i} = s'|_{S_i}$ for all i then $s = s'$,
- ② If there exists $s_i \in \mathcal{F}(S_i)$ for all i such that $s_i|_{S_i \cap S_j} = s_j|_{S_i \cap S_j}$, then there exist an $s \in \mathcal{F}(S)$ such that $s|_{S_i} = s_i$ for all i .

- It is enough for us to define a sheaf $\mathcal{F}$ over a basis of the topology of X .

Review: Basic Algebraic Geometry

- Let R be a ring.
- The **spectrum** of R is defined as $\text{Spec } R = \{\mathfrak{p} : \mathfrak{p} \text{ is a prime ideal of } R\}$, endowed with a topology such that the basic open sets are of the form $D(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}$, i.e. these open sets form a basis.
- For each basic open set $D(f) \subseteq \text{Spec } R$, we can define a ring $\mathcal{O}_{\text{Spec } R}(D(f)) = R_f$, where

$$R_f = \{f^n \mid n \geq 0\}^{-1}R = \left\{ \frac{a}{f^n} \mid a \in R, n \geq 0 \right\}.$$

- Thus we can define an **affine scheme** to be a pair $(X, \mathcal{O}_X)$, where $X = \text{Spec } R$ for some ring R , and $\mathcal{O}_X$ is defined for each open subset of X as above.
- Finally, we can define a **scheme** to be a pair $(X, \mathcal{O}_X)$ such that X has an open cover $(U_i)_i$, where each $(U_i, \mathcal{O}_X|_{U_i})$ are affine schemes.

Review: Basic Algebraic Geometry

- As an example, the projective plane $\mathbb{P}_k^2$, defined by gluing three affine planes $\mathbb{A}_k^2$ along the open sets $D(x)$.
- Geometrically, we can describe the projective plane as the three dimensional space where the "points" are equivalence classes of lines and the "curves" are "cones" that pass through $(0, 0, 0)$.
- Algebraically, $\mathbb{P}_k^2$ consists of three affine planes $U_\kappa = \mathbb{A}_k^2 = \text{Spec } k[X_{i\kappa}, X_{j\kappa}]$, such that $X_{i\kappa} = x_i/x_\kappa$ and $U_{\kappa j} = D_{U_\kappa}(x_j/x_\kappa) \subseteq U_\kappa$.
- Define morphisms $\varphi_{ji} : U_{ij} \rightarrow U_{ji}$ by the associated ring homomorphism $x_i/x_j \mapsto x_j/x_i$.
- We get a scheme X with morphisms $\psi_i : U_i \rightarrow X$ such that $\psi_i \circ \varphi_{ji} = \psi_j$ over U_{ij} , $X = \bigcup_i \psi_i(U_i)$, $\psi_i(U_i) \cap \psi_j(U_j) = \psi_i(U_{ij}) = \psi_j(U_{ij})$, and ψ_i gives an isomorphism from U_i to an open subscheme of X .
- This pair X and (ψ_i) are unique up to unique isomorphism, thus we can give X a name: $\mathbb{P}_k^2$.
- Constructions like these are called Proj, and in this case we write $\mathbb{P}_k^2 = \text{Proj}(k[X, Y, Z])$.

Schemes, Continued

Smoothness

- We can define smoothness using Jacobians.

Definition

Let $f : X \rightarrow Y$ be a morphism of schemes and $d \geq 0$ be an integer.

- ① We say that f is **smooth of relative dimension** d at a point $x \in X$ if there exists open affine neighborhoods U of x and $V = \operatorname{Spec} R$ of $f(x)$ such that $f(U) \subseteq V$ and an open immersion $U \hookrightarrow \operatorname{Spec} R[\underline{T}]/\langle f_1, \dots, f_{n-d} \rangle$ of R -schemes for some n and f_i such that the Jacobian

$$J_{f_1, \dots, f_{n-d}}(x) = \left(\frac{\partial f_i}{\partial T_j}(x) \right)_{i,j}$$

has rank $n - d$.

- ② We say that $f : X \rightarrow Y$ is **smooth** or X is **smooth** over Y if it is smooth at all points $x \in X$.

- As an example, if $X = V(y^2 - 4x^3 + x + 1) \subseteq \mathbb{A}_{\mathbb{Z}}^2$, then it is smooth of relative dimension 1 over $\mathbb{Z}$.

Curves

- Let k be a field (for example $\mathbb{Q}$, $\mathbb{F}_p$) and $f \in k[x, y]$.
- Classically, an affine **curve** γ over a field k is the zero set

$$\mathcal{C} = \{\mathfrak{p} \in \operatorname{Spec} k[x, y] \mid f \in \mathfrak{p}\} = V(f).$$

- More generally, for an ideal $\mathfrak{a}$ of $k[x, y]$, we define

$$\mathcal{C} = \{\mathfrak{p} \in \operatorname{Spec} k[x, y] \mid \mathfrak{a} \subseteq \mathfrak{p}\} = V(\mathfrak{a}).$$

- If $\mathfrak{a}$ is a prime ideal $\mathfrak{p}$, then the curve $V(\mathfrak{p})$ is irreducible.
- More specifically, if $\mathfrak{a} = \langle f \rangle$ for an irreducible polynomial f , then $V(\mathfrak{a}) = V(f)$ is irreducible.

Curves

- Let $\mathfrak{a} = \prod_{i=1}^n \mathfrak{p}_i = \{\sum \alpha f_1 f_2 \cdots f_n \mid f_i \in \mathfrak{p}_i\}$.
- By definition we have

$$V(\mathfrak{a}) = \bigcup_{\mathfrak{p}|\mathfrak{a}} V(\mathfrak{p}).$$

- Similarly to schemes, the plane curves $V(\mathfrak{p})$ is called an **irreducible component** of $V(\mathfrak{a})$.
- If the ideal $\bar{\mathfrak{p}}$ in $\bar{k}[X, Y]$ remains a prime ideal, the plane curve $V(\mathfrak{p})$ is called **geometrically irreducible**.

Projective Curves

- In the projective space $\mathbb{P}^2$, the definition is similar.
- For homogeneous ideals $\mathfrak{a}$, write the dehomogenization of $\mathfrak{a}$ by X_κ as $\mathfrak{a}_\kappa$ and

$$V_+(\mathfrak{a}) = \{\mathfrak{p} \in \text{Proj } k[X, Y, Z] \mid \mathfrak{a} \subseteq \mathfrak{p}\} = \bigcup \{\mathfrak{p} \in \text{Spec } k[X_i, X_j] \mid \mathfrak{a}_\kappa \subseteq \mathfrak{p}\}$$

Definition (Projective Plane Curve)

A **projective plane curve** is a closed subscheme $C = V_+(\mathfrak{a})$ of $\mathbb{P}_k^2$, where $\mathfrak{a} \subseteq k[X, Y, Z]$ is a homogeneous ideal. If $\mathfrak{a} = \langle f \rangle$, then the degree of f is called the **degree** of C .

- An easy example is a line: take $\mathfrak{a} = \langle f \rangle$, where $f = aX + bY + cZ$.
- We of course have another example that will be the main topic later, namely elliptic curves: take $\mathfrak{a} = \langle f \rangle$, where $f = Y^2Z - 4X^3 + g_2XZ^2 + g_3Z^3$.
- Going forward, we assume that all curves are projective plane curves.

Projective Curves

- Let C be an irreducible curve.
- The **rational function field** of C over k , written as $k(C)$, is defined to be $\text{Frac } \mathcal{O}_{C,P}$ for any $P \in C(\bar{k})$.
- As an example, if $C = \text{Proj } k[x, y] = \mathbb{P}_k^1$, the function field $k(C)$ is just $k(t)$ by taking standard affine open covers and calculating the fraction field of its stalks.
- We have the following lemma.

Lemma

Let $0 \neq f \in k(C)$. Then there exists homogeneous polynomials $G(X, Y, Z), H(X, Y, Z) \in k[X, Y, Z]$ with $\deg G = \deg H$ such that $f = H/G$.

- As a corollary, we have:

Corollary

For smooth curves C , each $\phi \in k(C)$ induces $\phi \in \text{Hom}_{\text{proj curves}}(C, \mathbb{P}^1)$.

Family of Curves

- We can generalize the definition of a curve as follows.

Definition (Family of Curves)

Let S be a scheme. An S -scheme $f : \mathcal{C} \rightarrow S$ is called a **family of curves** over S if f is a flat, projective morphism and all fibers of f are equidimensional of dimension 1.

- By "family" we mean, via base change to $\text{Spec } k$, $f : \mathcal{C} \rightarrow S$ generates a curve.

$$\begin{array}{ccc} \mathcal{C}_s & \longrightarrow & \mathcal{C} \\ \downarrow & & \downarrow \\ \text{Spec } \kappa(s) & \longrightarrow & S \end{array}$$

Figure 1: Family of Curves

Algebro-Geometric Toolbox

Cohomological Tools

- For this work, we only need the zeroth and first Čech cohomological groups, H^0 and H^1 .
- The usual definitions require some complexes, which are generalizations of exact sequences.
- However, for now, we shall use an easier characterization, given by the following.

Lemma

Let $\mathcal{F}$ be a sheaf on a scheme X and $\mathcal{U} = (U_i)$ is an affine open cover of $U \subseteq X$ open. Then

- ① $H^0(\mathcal{U}, \mathcal{F}) = \Gamma(U, \mathcal{F})$,
- ② $H^1(\mathcal{U}, \mathcal{F}) = \{(f_{ij}) \mid f_{ij} \in \mathcal{F}(U_i \cap U_j), f_{kj} + f_{ji} = f_{ki}, i < j < k\} / \{f_i - f_j \mid f_i \in \mathcal{F}(U_i)\}$.

- We shall see a justification of the second claim later on.
- As an example, if $X = V(y^2 - 4x^3 + x + 1)$ with itself as an open cover and $\mathcal{F} = \mathcal{O}_X$, then $H^1(X, \mathcal{F}) = \{f \in \mathcal{F}(X) = \mathcal{O}_X(X) \mid 2f = f\} \Rightarrow \{0\}$.

Divisor

- Let X be a scheme.
- We want to consider a configuration of roots and poles of a rational function over X .
- This is a natural consideration, since in complex analysis, a meromorphic function f is determined by its roots and poles up to a multiple of $e^{g(z)}$, where g is entire.
- To do this, there are two ways.
- Firstly, we can consider a formal $\mathbb{Z}$ -linear combination of irreducible closed subsets with codimension 1, which brings us to **Weil divisors**.
- Secondly, we can consider equivalence classes of rational functions, which brings us to **Cartier divisors**.

Divisor

- We start with Cartier divisors, which are more general.

Definition (Cartier Divisor)

Let X be a scheme.

- ① A **Cartier divisor** on X is an element of the abelian group $\text{Div}(X) = \Gamma(X, \mathcal{K}_X^\times / \mathcal{O}_X^\times)$.
- ② A divisor is called **principal** if it is in the map of the canonical homomorphism $\Gamma(X, \mathcal{K}_X^\times) \rightarrow \text{Div}(X)$. A principal divisor defined by f is written as $\text{div}(f)$.
- ③ Two divisors D_1, D_2 are **linearly equivalent** if $D_1 - D_2$ is principal.
- ④ A divisor D is called **effective** if it is in the submonoid $\text{Div}_+(X) = \Gamma(X, (\mathcal{K}_X^\times \cap \mathcal{O}_X) / \mathcal{O}_X^\times)$, written as $D \geq 0$. For two divisors D, D' , we write $D \geq D'$ if $D - D' \geq 0$.

Divisor

- We can describe Cartier divisors more concretely.
- For $f \in \Gamma(U, \mathcal{K}_X)$ and $g \in \Gamma(V, \mathcal{K}_X)$, write fg for $f|_{U \cap V} g|_{U \cap V}$.
- A Cartier divisor is given by a tuple $(U_i, f_i)_i$, with $(U_i)_i$ is an open cover of X and $f_i \in \Gamma(U_i, \mathcal{K}_X^\times)$ is a meromorphic function such that $f_i f_j^{-1} \in \Gamma(U_i \cap U_j, \mathcal{O}_X^\times)$ for all i, j .
- For divisors $(U_i, f_i)_i, (V_j, g_j)_j$, these two divisors are equal iff $f_i g_j^{-1} \in \Gamma(U_i \cap V_j, \mathcal{O}_X^\times)$ for all i, j .
- The sum of two divisors is given as $(U_i \cap V_j, f_i g_j)_{i,j}$, and the inverse of a divisor is given as $(U_i, f_i^{-1})_i$.
- A divisor is effective iff $f_i \in \Gamma(U_i, \mathcal{O}_X)$ for all i .
- A divisor is principal if it is given by (X, f) .

Divisor

- To illustrate this definition, let $f, g : \mathbb{C} \rightarrow \mathbb{C}$ be meromorphic functions.
- If $f = ug$ for some unit $u = e^{h(z)}$, then it is natural that f and g have the same configuration of zeros and poles.
- This characterization, for **entire** complex functions, is called **Hadamard's Factorization Theorem**.
- We can recover this characterization for meromorphic functions by dividing two entire functions.
- Thus, up to some unit multiple, we can write the configuration of roots and poles of a meromorphic function f as $\sum \lambda_i [z_i] - \sum \mu_j [w_j]$ for positive λ_i, μ_j .

Divisor

- Now we define Weil divisors.
- First define the **codimension** of $C \subseteq X$ as $\text{codim}_X C = \dim \mathcal{O}_{X,C}$.
- For $i \geq 0$ an integer, write the set of closed integral subschemes of codimension i by X^i .
- The free abelian group $\mathbb{Z}^{(X^i)}$ is denoted by $Z^i(X)$, with elements of the form $\sum_C n_C [C]$.

Definition (Weil Divisor)

A **Weil divisor** is an element of $Z^1(X)$. The elements of X^1 are called **prime Weil divisors**. Weil divisors of the form $\sum_{C \in X^1} n_C [C]$ such that $n_C \geq 0$ for all C are called **effective Weil divisors**, the set of which is denoted by $Z_+^1(X)$.

Divisor

- We can relate Cartier divisors and Weil divisors as follows.

Lemma

We have a group homomorphism $\text{cyc} : \text{Div}(X) \rightarrow Z^1(X)$ as $\text{cyc}(D) = \sum_{C \in X^1} \text{ord}_C(D)[C]$, where $\text{ord}_C(D) = \text{ord}_{\mathcal{O}_{X,C}}(f) = v_C(f)$.

- If X is a noetherian locally factorial scheme, then cyc is an isomorphism.
- Here, locally factorial means:

Definition (Locally Factorial)

A scheme X is called **locally factorial** if for all $x \in X$, the local ring $\mathcal{O}_{X,x}$ is a unique factorization domain.

Applications: Jacobian

- We can define special groups relating with divisors.
- Let $\text{Div}^0(X)$ be a subgroup of $\text{Div}(X)$ that consists of divisors D of degree 0.
- Inside that group, we have another subgroup $\{\text{div } f : f \in \mathcal{K}(X)^\times\}$.

Definition

The quotient groups

$$\mathcal{J}(X) = \text{Div}^0(X) / \{\text{div } f : f \in \mathcal{K}(X)^\times\}, \quad \text{Pic}(X) = \text{Div}(X) / \{\text{div } f : f \in \mathcal{K}(X)^\times\}$$

are called the **Jacobian** and the **Picard group** of X respectively.

- For example, if $X = \mathbb{P}_k^1$, then as we shall see later, $\text{Pic}(X) \simeq \mathbb{Z}$ and $\mathcal{J}(X) \simeq \{1\}$.

Connection with Line Bundles

- Here we attach a divisor D of a scheme X to a sheaf $\mathcal{O}_X(D)$ such that for all $U \subseteq X$ open, it is an $\mathcal{O}_X(U)$ -submodule of $\mathcal{K}_X(U)$.
- This is called a **line bundle**.
- Firstly we define the notion of **invertible fractional ideal**.

Definition

An **invertible fractional ideal** is an invertible $\mathcal{O}_X$ -submodule of $\mathcal{K}_X$.

Definition

Let D be a divisor on X , represented by $(U_i, f_i)_i$. There exists a unique invertible fractional ideal $\mathcal{I}_X(D)$ such that $\mathcal{I}_X(D)|_{U_i} = \mathcal{O}_{U_i}(f_i)$. Define the line bundle attached to D as $\mathcal{L}(D) = \mathcal{O}_X(D) = \mathcal{I}_X(D)^{-1}$. The notations $\mathcal{O}_X(D)$ and $\mathcal{L}(D)$ shall be used interchangeably.

Connection with Line Bundles

- Concretely,

$$\Gamma(U, \mathcal{O}_X(D)) = \{f \in \mathcal{K}_X(U) : f\mathcal{I}_X(D)(U) \subseteq \mathcal{O}_X(U)\}.$$

- The global sections of the line bundle $\mathcal{O}_X(D)$ can be written as

$$\Gamma(X, \mathcal{O}_X(D)) = \{f \in \mathcal{K}_X^\times(X) : \operatorname{div} f + D \geq 0\} \cup \{0\}.$$

- All line bundles that we consider are of this form.

Line Bundles

- More generally,

Definition (Line Bundle)

A **line bundle** is a sheaf $\mathcal{I}$ over X such that for every open set $U \subseteq X$, $\mathcal{I}(U)$ is an $\mathcal{O}_X(U)$ -module that is free of rank 1.

- A module of rank 1 is a generalization of vector spaces of dimension 1, but the base field is replaced by a (commutative) ring.
- We can collect all isomorphism classes of line bundles over X and give it an operation $\otimes$ defined as taking the tensor product, which gives it an Abelian group structure.
- This Abelian group is called the **Picard group** of the scheme X .

Line Bundles

- Let $\mathcal{F}$, $\mathcal{G}$, and $\mathcal{H}$ be sheaves over a scheme X such that the following sequence is exact:

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$$

- Taking Čech cohomology, we get another exact sequence

$$0 \rightarrow H^0(X, \mathcal{F}) \rightarrow H^0(X, \mathcal{G}) \rightarrow H^0(X, \mathcal{H}) \rightarrow H^1(X, \mathcal{F}) \rightarrow \dots$$

- Thus $H^0(X, \mathcal{H}) = \{(g_i, U_i) : g_i \in \mathcal{G}(U_i), g_i - g_j \in \mathcal{F}(U_i \cap U_j)\}$.
- Thus the mapping $H^0(X, \mathcal{H}) \rightarrow H^1(X, \mathcal{F})$ is defined as $(g_i, U_i) \mapsto ((g_i - g_j)_{i < j}, U_i)$, which means $H^1(X, \mathcal{F}) = \{(f_{ij}) : f_{ij} \in \mathcal{F}(U_i \cap U_j), f_{ij} - f_{jk} + f_{ki} = 0\}$.
- From this we can describe vector bundles as the first Čech cohomology.

Line Bundles

- Let $\mathbb{P}_k^1$ be the projective line over a field k .
- We can define a line bundle $\mathcal{O}_{\mathbb{P}_k^1}(n)$ for any integer n as follows.
- First, take the standard affine open cover of $\mathbb{P}_k^1$, write the two affine lines as U_0 and U_1 .
- We take U_0, U_1 as the trivializer of the line bundle $\mathcal{O}_{\mathbb{P}_k^1}(n)$.
- Define a global section (α_i) of this line bundle such that $\alpha_i \in \mathcal{O}_{\mathbb{P}_k^1}(U_i) \simeq k[X_i]$, in $U_0 \cap U_1$, $\alpha_0 - \alpha_1 = 0$, and α_0, α_1 are related by transition functions:
 $\alpha_0(X_0) = X_0^n \alpha_1(1/X_0^n)$.
- Thus $\alpha_0(X_0) = a_0 + a_1 X_0^1 + \cdots + a_n X_0^n$, similarly for $\alpha_1(X_1)$.
- The integer n corresponds to the "twisting" of each of the lines.

Differential

- An important example of line bundles on curves are **differential sheaves**.
- Let R be a ring and A be an R -algebra.
- To illustrate what we want, let $R = k$ and $A = k[x, y]/\langle y^2 - x^3 - x \rangle$, thus we can define a module of differentials $\Omega_{A/R}$ as $Adx \oplus Ady$ subject to the relation $2ydy - (3x^2 + 1)dx = 0$.
- To do that, first we define a mapping $d : A \rightarrow \Omega_{A/R}$ as an R -linear map such that $d(xy) = ydx + xdy$ for $x, y \in A$, and for all $r \in R$, $dr = 0$.
- For $A = R[x_1, \dots, x_n]/I$, we define our differential module, $\Omega_{A/R}$, to be $(Adx_1 \oplus \dots \oplus Adx_n)/\{dr \mid r \in I\}$, and for $A = R/I$, we define $\Omega_{A/R} = 0$.

Differential

- Thus for affine schemes $\text{Spec } A \rightarrow \text{Spec } R$, the differential sheaf is defined as $\widetilde{\Omega_{A/R}}$, i.e. for all basic open sets $D(f)$ of $\text{Spec } A$, we must have $\widetilde{\Omega_{A/R}}(D(f)) = \Omega_{(A/R),f} = \Omega_{A_f/R}$ (which we shall write without the tilde).
- Let $f : X \rightarrow Y$ be a scheme, which for this seminar we assume that $Y = \text{Spec } R$ is affine.
- Per definition, X has an affine open cover $(U_i) = (\text{Spec } A_i)$.
- Thus we can define the differential sheaf of X , written as $\Omega_{X/Y}$, by defining $\Omega_{X/Y}(U_i) = \Omega_{A_i/R}(U_i)$.

Riemann-Roch

- As an application of differentials, we have the following.

Theorem (Riemann-Roch)

Let C be a smooth projective irreducible curve over a field k . Let $K = \operatorname{div} \omega$ for a meromorphic differential $0 \neq \omega \in \Omega_{C/\bar{k}, \eta}$. For $g = \dim_{\bar{k}} \Omega_{C/\bar{k}}$, we have

$$\ell(D) = 1 - g + \deg(D) - \ell(K - D)$$

for all divisors D on $C(\bar{k})$.

- Here, $\ell(D) = \dim_{\bar{k}} \mathcal{L}(D)$.
- The number g is called the **geometric genus** of the curve C .

Elliptic Curves

Elliptic Curves

- Now we are ready to define elliptic curves over fields.
- Classically, we define elliptic curves E over k as the roots of the polynomial $y^2 = 4x^3 - g_2x - g_3$ along with a point at infinity, which we shall see later is actually a consequence of the following definition.

Definition (Elliptic Curves over Fields)

An **elliptic curve** $E \rightarrow \operatorname{Spec} k$, written as E/k , is a projective plane curve that is irreducible and smooth with genus 1 and a distinguished point 0_E .

- Classically, we take $0_E = [0 : 1 : 0]$.
- The following theorem gives elliptic curves the structure of an Abelian group.

Theorem (Abel)

Let E/k be an elliptic curve with distinguished point 0_E . The correspondence $P \mapsto [P] - [0_E]$ induces a bijection $E(\bar{k}) \xrightarrow{\sim} \mathcal{J}(E)$. Thus $E(\bar{k})$ is an abelian group.

Elliptic Curves

- As a corollary to Abel's Theorem, we have the following.

Corollary

If $0 \neq \omega \in \Omega_{E/\bar{k}}$, then $\operatorname{div}(\omega) = 0$.

- Thus the differential ω vanishes nowhere, which from now on we shall call a **nowhere vanishing differential**.
- Moreover, by the definition of genus, we have that the only nowhere vanishing differential in $\Omega_{E/\bar{k}}$ is ω up to constant multiple.
- Thus, nowhere vanishing differentials characterizes elliptic curves.

Elliptic Curves

- Pressingly, we don't know how to define an elliptic curve as a vanishing scheme.
- Assume that $\text{char } k \neq 2, 3$.

Proposition

Let (E, ω) be a pair of an elliptic curve E and a nowhere vanishing differential ω both defined over k . There exists a unique basis $\{1, x, y\}$ of $\mathcal{L}(3[0_E])$ such that E can be embedded into $\mathbb{P}_k^2$ by $\{1, x, y\}$ whose map is defined as the affine equation

$$y^2 = 4x^3 - g_2x - g_3 \text{ with } g_2, g_3 \in k$$

and the map of ω is given by dx/y . Conversely, a projective algebraic curve that is defined by the affine equation is an elliptic curve with nowhere vanishing differential dx/y iff $\Delta(E, \omega) = g_2^3 - 27g_3^2$ is nowhere vanishing.

Elliptic Curves

- By the previous proposition, we can define an elliptic curve as an affine equation.
- For example, $y^2 = 4x^3 - x - 1$ defines an elliptic curve, since $\Delta(E, \omega) = -1 - 27 = -28 \neq 0$.

Definition

The function $\Delta(E, \omega)$ is called the **discriminant function** or **Ramanujan's Δ -function**.

- The Δ -function characterizes the nature of the roots of the cubic $f = 4x^3 - g_2x - g_3$: it is 0 if f has a double root and nonzero otherwise.
- Specifically, the Δ -function is a special case of the discriminant function for a cubic.

Elliptic Curves

Proof Sketch.

We construct the equations directly by considering the global sections of the line bundle $\mathcal{L}(6[0_E])$, constructing a basis for it by considering the line bundles $\mathcal{L}([0_E])$ to $\mathcal{L}(3[0_E])$, which we write:

$$\begin{aligned}\mathcal{L}([0_E]) &= k, \mathcal{L}(2[0_E]) = k + kx, \mathcal{L}(3[0_E]) = k + kx + ky, \\ \mathcal{L}(4[0_E]) &= k + kx + ky + kx^2, \mathcal{L}(5[0_E]) = k + kx + ky + kx^2 + kxy, \\ \mathcal{L}(6[0_E]) &= k + kx + ky + kx^2 + kxy + kx^3 = k + kx + ky + kx^2 + kxy + ky^2.\end{aligned}$$

Remember that these line bundles have global sections that can be thought of as functions that can have a pole of order n at 0_E . Also recall that an elliptic curve cannot have a rational function with a pole of order 1. Thus after a few changes of the variables, we get the equation.

The nowhere vanishing differential ω is defined as dx/y . □

Elliptic Curves

- Previously we stated that $E(\bar{k})$ is an Abelian group.
- Now we define an operation that can be easily computed (there is an algorithm for this, but it is not within the scope of this work).

Lemma

Suppose that a line $\ell : ax + by + cz = 0$ in $\mathbb{P}^2$ intersects an elliptic curve E at P, Q, R . Then $[P] + [Q] + [R] \sim 3[0_E]$.

- Thus, constructing a group operation in an elliptic curve can be done geometrically.
- An illustration is provided in the next slide.

Elliptic Curves

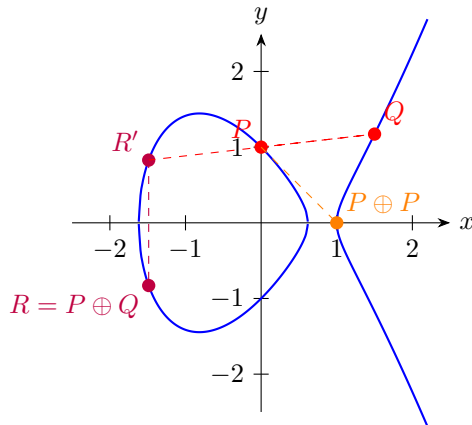


Figure 2: Group Operation in an Elliptic Curve

Elliptic Curves

- Via abelian schemes, we can define the following, and later we shall see that this generalizes the definition of elliptic curves over k .

Definition (Families of Elliptic Curves)

Let S be a scheme. A smooth projective curve $f : E \rightarrow S$ is a **family of elliptic curve** if it satisfies:

- ① E has a section $0 = 0_E \in E(S)$, called a **zero section** such that $f \circ 0 = 1_S$.
- ② $\dim_S E = 1$ and E is geometrically irreducible fiber by fiber, that means each geometric fiber of E over an algebraically closed field is smooth, irreducible, and of dimension 1.
- ③ For the $\mathcal{O}_E$ -module of 1-differentials $\Omega_{E/S}$, $f_*\Omega_{E/S}$ is locally free of rank 1.

- As an example, consider the curve $E \rightarrow \operatorname{Spec} \mathbb{C}[t]$ "defined" by the following equation (we shall justify this later, when we talk about moduli)

$$y^2 = 4x^3 - tx - 1.$$

- As long as $t^3 - 27 \neq 0$, this defines an elliptic curve.

Moduli of Elliptic Curves

Moduli?

- The main theme of this section is as follows.
- Suppose that we have a functor $\varphi : (\text{Something}) \rightarrow (\text{Sets})$, defined as

some object $\mapsto$ {some similar objects up to isomorphism}.

- A natural question: is this functor representable by some object (usually a scheme)?
- If it is, then we call the representation a **fine moduli space**.
- If it is not, but we can find an object (usually a scheme) that closely represents it, we call the object a **coarse moduli space**.
- The theory is rich, thus sadly, we cannot talk about it in detail, but we shall see some examples.

Moduli of Weierstrass Type

- First, consider an elliptic curve over a field E/k with nowhere vanishing differential ω .
- We want to characterize these elliptic curves, up to isomorphism.

Definition

Two pairs of elliptic curves $(E, \omega), (E', \omega')$ over k is said to be **isomorphic** if there exists an isomorphism $\varphi : E \rightarrow E'$ with $\varphi^*\omega' = \omega$ and $\varphi(0_E) = 0_{E'}$.

- From this definition, we can see that by choosing some local parameter T, T' at $0_E, 0_{E'}$, E and E' has the same Weierstrass equation.

Moduli of Weierstrass Type

- Define a classification functor $\wp(k) = \{(E, \omega)/k\} / \sim$, where $(E, \omega) \sim (E', \omega')$ if they are isomorphic.
- Then we have

$$\begin{aligned} \wp(k) &\simeq \{(g_2, g_3) \in k^2 : \Delta(E, \omega) \neq 0\} \\ &\simeq \operatorname{Hom}_{(\mathbb{Z}\text{-Alg})} \left(\mathbb{Z} \left[\frac{1}{6}, x, y, \frac{1}{x^3 - 27y^2} \right], k \right). \end{aligned}$$

- This is our first moduli result: the functor $\wp$ is represented by the scheme $\operatorname{Spec} \mathbb{Z} \left[\frac{1}{6}, x, y, \frac{1}{x^3 - 27y^2} \right]$.

Moduli of Weierstrass Type

- Let k be a field with $\text{char } k \neq 2, 3$.
- We now classify elliptic curves E over fields k (note the lack of the invariant differential ω).
- We have the following result.

Theorem

Let $E/k, E'/k$ be two elliptic curves that are isomorphic, with nowhere vanishing differentials $\omega_E, \omega_{E'}$ respectively. Then $g_j(E', \omega_{E'}) = \lambda^{-2j} g_j(E, \omega_E)$ for $\lambda \in k^\times$ such that $\varphi^ \omega_{E'} = \lambda \omega_E$.*

Moduli of Weierstrass Type

- This classification goes even further: assuming k is algebraically closed, we can define a number J , depending on E , such that $J(E)$ classifies E .

Definition (J -Invariant)

Define the J -invariant of E as

$$J(E) = \frac{(12g_3(E, \omega))^3}{\Delta(E, \omega)}.$$

Corollary

If k is algebraically closed, then $J(E) = J(E') \Leftrightarrow E \simeq E'$. Moreover, for any field k , given $j \in k$, there exists an elliptic curve E with J -invariant $J(E) = j$.

Modular Forms

- Now we shall define a complex-analytic object that does not seem to relate to elliptic curves at all.
- However, we shall see later that this object is half of the key of the modularity theorem.

Definition (Weak Modularity)

Let $n \in \mathbb{Z}$. A meromorphic function $f : \mathbb{H} \rightarrow \mathbb{C}$ is called **weakly modular of weight n** if

$$f(\gamma(\tau)) = (c\tau + d)^n f(\tau) \text{ for } \gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}_2(\mathbb{Z}) \text{ and } \tau \in \mathbb{H}.$$

Modular Forms

Definition (Modular Forms, Analytic)

Let $n \in \mathbb{Z}$. A function $f : \mathbb{H} \rightarrow \mathbb{C}$ is called a **modular form with weight n** if

- ① f is holomorphic in $\mathbb{H}$,
- ② f is weakly modular of weight n ,
- ③ f is holomorphic at ∞ .

The set of all modular forms of weight n shall be denoted as $\mathcal{M}_n(\mathrm{SL}_2(\mathbb{Z}))$.

Modular Forms

- Now we shall define modular forms more geometrically.
- Firstly we need to define a generalization of the previous classification functor $\wp$.
- Let R be a $\mathbb{Z}[1/6]$ -algebra.
- Define the functor $\wp_R : (R\text{-Alg}) \rightarrow (\text{Sets})$ as

$$\wp_R(-) = \{(E, \omega)/-\}/\simeq.$$

- This functor can be represented in a similar manner, as we shall see in the next slide.

Modular Forms

- Let E be an elliptic curve over a $\mathbb{Z}[1/6]$ -algebra R .

Lemma

For $\mathcal{R}_R = \mathbb{Z}[g_2, g_3, 1/\Delta]$, the functor $\wp_R$ can be represented by $\mathcal{R}_R$, as in

$$\wp_R(-) = \mathrm{Hom}_{(R\text{-Alg})}(\mathcal{R}_R, -).$$

Theorem

Every morphism of functors $\varphi : \wp_R \rightarrow \mathbb{A}_R^1$ can be written as rational functions $\Phi \in \mathcal{R}_R$ over g_2 and g_3 such that $\varphi(E, \omega) = \Phi(g_2(E, \omega), g_3(E, \omega))$ for all elliptic curves (E, ω) over some R -algebra.

Modular Forms

- Define a **weight** function $w : R[g_2, g_3] \rightarrow \mathbb{Z}$ as $w(g_2^a g_3^b) = 4a + 6b$.
- For polynomials of the form $\Phi = \sum_{a,b} c_{a,b} g_2^a g_3^b$, define $w(\Phi) = \max\{w(g_2^a g_3^b) : c_{a,b} \neq 0\}$.
- Polynomials of this form is said to be **isobaric** if $c_{a,b} \neq 0$ implies $w(g_2^a g_3^b) = w$ for a constant w .
- With all these in hand, we can now define **geometric modular forms**.
- This definition is extremely long and shall be relegated to the next slide.

Modular Forms

Definition (Modular Forms, Geometric)

A **modular form** of weight w defined over R is a morphism of functors $\wp_R \rightarrow \mathbb{A}_R^2$ given by an isobaric polynomial over g_2 and g_3 of weight w and with coefficients in R . Write the R -module of modular forms of weight w as $\mathcal{M}_w(R)$. Thus $f \in \mathcal{M}_w(R)$ is a functorial rule, mapping isomorphism classes $(E, \omega)/B$ for some R -algebra B to $f(E, \omega) \in B$ such that

- ① $f \in R[g_2, g_3]$.
- ① If (E, ω) is defined over an R -algebra B , $f(E, \omega) \in B$ only depends on the isomorphism class of (E, ω) in B .
- ② $f((E, \omega) \otimes_B B') = \sigma(f(E, \omega))$ for all R -algebra homomorphisms $\sigma : B \rightarrow B'$.
- ③ $f((E, \lambda\omega)/B) = \lambda^{-w} f(E, \omega)$ for all $\lambda \in B^\times$.

- This definition gives an explicit algebraic definition for **level 1** modular forms.
- These two definitions are related by q -expansion.

Analytic Modular Forms

- Let $L \in \text{Lat}_{\mathbb{C}}$.

Definition (Weierstrass $\wp$ -Function)

Define

$$x_L(u) = \wp(u) = \frac{1}{u^2} + \sum_{\ell \in L \setminus \{0\}} \left(\frac{1}{(u - \ell)^2} - \frac{1}{\ell^2} \right),$$
$$y_L(u) = \wp'(u) = -\frac{2}{u^3} - 2 \sum_{\ell \in L \setminus \{0\}} \frac{1}{(u - \ell)^3}.$$

- Define

$$g_2 = g_2(L) = 60 \sum_{\ell \in L \setminus \{0\}} \frac{1}{\ell^4}, \quad g_3 = g_3(L) = 140 \sum_{\ell \in L \setminus \{0\}} \frac{1}{\ell^6}.$$

Analytic Modular Forms

- By equating poles, $\varphi = y_L^2 - 4x_L^3 + g_2x_L + g_3$ is holomorphic everywhere.
- Since these functions factor through $\mathbb{C}/L$, φ must be bounded and thus constant.
- Since x_L and y_L has constant term 0, $\varphi = 0$.
- This, along with Archimedean uniformization, proves the following theorem.

Theorem

We have $\wp(\mathbb{C}) = \{(E, \omega)/\mathbb{C}\}/\sim \simeq \text{Lat}_{\mathbb{C}}$.

- This is another "moduli" result: this theorem classifies all elliptic curves over $\mathbb{C}$ and relates them bijectively to lattices over $\mathbb{C}$.
- Moreover, by Archimedean uniformization, we see that every elliptic curve over $\mathbb{C}$ is modular.
- The special curve defined by φ , if g_2, g_3 have coefficients from $\mathbb{Z}[1/6]$, is called the **Tate curve**, written as $\text{Tate}(q)$.

Moduli of Families of Elliptic Curves

- There are many more classification/moduli problems concerning families of elliptic curves.
- We shall see a few examples that are... easier, although all of them are hard.
- This is the first example.

Theorem

Define a ring $\mathcal{R}_{\Gamma_1(1)} = \mathbb{Z}[1/6, g_2, g_3, 1/\Delta]$. The functor

$$\begin{aligned}\wp_{\Gamma_1(1)} : (\text{Ring}) &\rightarrow (\text{Sets}) \\ R &\mapsto \{(E, \omega)/R\}/\sim\end{aligned}$$

is representable by the fine moduli space $\mathcal{M}_1 = \text{Spec } \mathcal{R}_{\Gamma_1(1)}$.

- We shall only sketch the proof of these theorems in this seminar.

Moduli of Families of Elliptic Curves

Proof Sketch.

Let R be a ring. Locally we construct the Weierstrass equation in a similar manner, by considering the sections of the line bundle $\mathcal{L}(6[0_E])$. By gluing, we get an elliptic curve $(E, 0_E)$.

The nowhere vanishing differential ω is constructed similarly. □

- In addition to the previous theorem, we have the universal elliptic curve and the universal differential given by

$$(E_U, \omega_U)/\mathcal{M}_1 = \left(\text{Proj}(\mathcal{R}_{\Gamma_1(1)}[X, Y, Z]/\langle ZY^2 - 4X^3 + g_2XZ^2 + g_3Z^3 \rangle), \frac{dX}{Y} \right).$$

- For each pair $(E, \omega)/S$, we have a unique morphism $\varphi : S \rightarrow \mathcal{M}_1$ such that

$$(E, \omega)/S \simeq (E_U, \omega_U) \times_{\mathcal{M}_1} S.$$

Moduli of Families of Elliptic Curves

- The previous result can be generalized.

Theorem

The functor

$$\begin{aligned} \wp_{\Gamma_1(N)} : (\text{Ring}) &\rightarrow (\text{Sets}) \\ R &\mapsto \{(E, \phi_N : \mu_N \hookrightarrow E[N], \omega)/R\}/\sim \end{aligned}$$

is representable by the fine moduli space $\mathcal{M}_{\Gamma_1(N)} = \text{Spec } \mathcal{R}_{\Gamma_1(N)}$ for a ring $\mathcal{R}_{\Gamma_1(N)}$.

Proof Sketch.

For a prime p , consider the triple $(E, P, \omega)/R$ where P is a point of order p . By the universal elliptic curve, we have a unique morphism φ such that

$\varphi_E : (E, \omega)/R \xrightarrow{\sim} (E_U, \omega_U) \times_{\mathcal{M}_1} \text{Spec } R$. Thus P induces a unique morphism $\varphi_P = \varphi_E \circ P$, thus $\wp_{\Gamma_1(p)}(R) \simeq (E_U[p] \setminus \{0\})(R)$. For nonprime N , by a similar argument we have $\wp_{\Gamma_1(N)}(R) \simeq (E_U[N] \setminus \bigcup_{N>d|N} E_U[d])(R)$. □

Modular Forms

- As a generalization for the previous definition of geometric modular forms, we shall use the level structure given by $\Gamma_1(N)$ on the classification functor $\wp_{\Gamma_1(N)}$.
- Let B be a $\mathbb{Z}[1/(6N)]$ -algebra, and write $\wp_{\Gamma_1(N)/B}$ as the restriction of $\wp_{\Gamma_1(N)}$ to $(B\text{-Alg})$.
- We can split $\mathcal{R}_{\Gamma_1(N)/B} = \bigoplus_{\kappa \in \mathbb{Z}} \mathcal{R}_{\kappa}(\Gamma_1(N), B)$.
- Thus we can define modular forms with level N as follows.

Definition (Geometric Modular Forms)

Let $f \in \mathcal{R}_{\kappa}(\Gamma_1(N), B)$, which can be regarded as a function of $(E, \phi_N, \omega)/R$ such that $f((E, \phi_N, \omega)/R) \in R$ and

- ① $f((E, \phi_N, \lambda\omega)/R) = \lambda^{-\kappa} f((E, \phi_N, \omega)/R)$ for $\lambda \in R^{\times}$,
- ② $(E, \phi_N, \omega) \simeq (E', \phi'_N, \omega') \Rightarrow f((E, \phi_N, \omega)/R) = f((E', \phi'_N, \omega')/R)$,
- ③ if $\rho: R \rightarrow R'$ is a B -algebra homomorphism, then $f((E, \phi_N, \omega)/R \times_R R') = \rho(f((E, \phi_N, \omega)/R))$.

Hecke Operators

- We can define a special operator, called the **Hecke operator**, taking modular forms to another mapping.

Definition

For n relatively prime to N , define the **Hecke operator** $T(n)$ taking a modular form f to another map

$$f|T(n) : \mathcal{O}_{\Gamma_1(N)/B} \rightarrow \mathbb{G}_a/B,$$
$$(E, \phi_N, \omega) \mapsto \frac{1}{p} \sum_{C \in \mathcal{C}(n)} f(E/C, \pi_{C,*}\phi_N, \phi_{C,*}\omega).$$

- These operators are linear, and thus we can define an algebra, called the **Hecke algebra**, consisting of all Hecke operators over B .

Hecke Operators

- Classically, modular forms can be written as a Fourier expansion, which we then write $q = e^{2\pi iz/n}$ for its variable.
- We shall construct these q -expansions algebraically.

Definition

For every $\mathbb{Z}[1/(6N)]$ -algebra B , define $G_\kappa(\Gamma, B)$ to be the κ -th graded component of the ring $G_\Gamma(\mathbb{Z}[1/(6N)]) \otimes B$, where $\Gamma = \Gamma_1(N)$.

- Here, $G_\Gamma(\mathbb{Z}[1/(6N)])$ is the integral closure of the ring of modular forms over $\mathrm{SL}_2(\mathbb{Z})$ with level 1.
- Thus (via insane blackboxing measures to ensure I can finish this seminar in time) every $f \in G_\kappa(\Gamma_1(N), B)$ is a function of (E, ϕ_N, ω) and a modular form such that $f(\mathrm{Tate}(q), \phi_N, \omega) \in B[\zeta_N][[q^{1/N}]]$.
- In particular, for $B = \mathbb{C}$, $f(\mathrm{Tate}(q), \phi_N, \omega) = \sum_{n=0}^{\infty} a(n, f)q^n$ coincides with the Fourier expansion of f at infinity.

Ending

Conclusion

In these two seminars,

- We have seen the basic framework for modern algebraic geometry, namely **schemes**, **fiber products**, **line bundles**, **differential sheaves**, and as an application, **curves**.
- We have seen a generalization of the definition of elliptic curves, which we saw was a family of elliptic curves.
- We have seen that the Weierstrass equation of elliptic curves that are usually used to define them as varieties can be constructed from our generalized definition, and thus the properties of elliptic curves as varieties also apply for our generalized definition.
- We have seen that the role of schemes in the proof of the Modularity Theorem are as follows: as a tool to define families of elliptic curves and as a tool to define modular forms for families of elliptic curves.

Suggestions for Future Work

- There are a lot of things that we only discuss briefly either in this seminar or in the written thesis, such as complex multiplication, Galois deformations, and so on, that are interesting in its own right. However, some of them require things beyond scheme theory, for example algebraic spaces, stacks, and so on.
- There are generalizations of the Modularity Theorem for general Abelian schemes, for elliptic curves over number fields, and so on, that were too far to reach for this work, that can be discussed in further work.

References I

-  Boxer, G. et al. (2025). *Modularity theorems for abelian surfaces*. arXiv: 2502.20645 [math.NT].
-  Breuil, C. et al. (2001). “On the modularity of elliptic curves over $\mathbb{Q}$: Wild 3-adic exercises”. In: *Journal of the American Mathematical Society* 14.4, pp. 843–939. ISSN: 0894-0347. DOI: 10.1090/s0894-0347-01-00370-8.
-  Conrad, B., F. Diamond, and R. Taylor (1999). “Modularity of certain potentially Barsotti-Tate Galois representations”. In: *Journal of the American Mathematical Society* 12.2, pp. 521–567. ISSN: 0894-0347. DOI: 10.1090/s0894-0347-99-00287-8.
-  Crespo, T. and Z. Hajto (2011). *Algebraic Groups and Differential Galois Theory*. ISBN: 9781470411831.
-  Diamond, F. and J. Shurman (2006). *A First Course in Modular Forms*. Springer Science & Business Media. ISBN: 9780387272269.
-  Görtz, U. and T. Wedhorn (2020). *Algebraic Geometry I: Schemes*. Springer Nature. ISBN: 9783658307332.

References II

-  Görtz, U. and T. Wedhorn (2023). *Algebraic Geometry II: Cohomology of Schemes*. Springer Nature. ISBN: 9783658430313.
-  Guitart, X. and J. Quer (2014). “Modular Abelian Varieties Over Number Fields”. In: *Canadian Journal of Mathematics* 66.1, 170–196. DOI: 10.4153/CJM-2012-040-2.
-  Hida, H. (2011). *Geometric Modular Forms And Elliptic Curves (2nd Edition)*. World Scientific. ISBN: 9789814405232.
-  — (2013). *Elliptic Curves and Arithmetic Invariants*. Springer Science & Business Media. ISBN: 9781461466574.
-  Kleiman, S. and A. Altman (2013). *A Term of Commutative Algebra*. ISBN: 9780988557215.
-  Milne, J. S. (2008). *Abelian Varieties (v2.00)*. Available at www.jmilne.org/math/.
-  Milne, James S. (2017). *Modular Functions and Modular Forms (v1.31)*. Available at www.jmilne.org/math/.
-  Mumford, D. (2008). *Abelian Varieties*. Debolsillo. ISBN: 9788185931869.

References III

-  Munkres, J. R. (2000). *Topology*. Prentice Hall.
-  Neukirch, J. (2013). *Algebraic Number Theory*. Springer Science & Business Media. ISBN: 9783662039830.
-  nLab authors (Dec. 2025). *geometric fibre*.
<https://ncatlab.org/nlab/show/geometric+fibre>. Revision 4.
-  Reid, M. (1988). *Undergraduate Algebraic Geometry*. Cambridge University Press. ISBN: 9780521356626.
-  Ribet, K. A. (Dec. 1990). “On modular representations of $\bar{\mathbb{Q}}/\mathbb{Q}$ arising from modular forms”. In: *Inventiones Mathematicae* 100.1, pp. 431–476. ISSN: 0020-9910. DOI: 10.1007/bf01231195.
-  Rotman, J. J. (2008). *An Introduction to Homological Algebra*. Springer Science & Business Media. ISBN: 9780387683249.
-  Silverman, J. H. (2009). *The Arithmetic of Elliptic Curves*. Springer Science & Business Media. ISBN: 9780387094946.

References IV

-  Taylor, R. and A. Wiles (1995). “Ring-Theoretic Properties of Certain Hecke Algebras”. In: *Annals of Mathematics* 141.3, pp. 553–572. ISSN: 0003486X, 19398980. (Visited on 11/07/2025).
-  Wiles, A. (1995). “Modular Elliptic Curves and Fermat’s Last Theorem”. In: *Annals of Mathematics* 141.3, pp. 443–551. ISSN: 0003486X, 19398980. (Visited on 11/07/2025).

Thanks for your attention.
Any questions?